INCH-POUND

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DETAIL SPECIFICATION

INFLATION DEVICE, AUTOMATIC, FLU-8B/P AND FLU-9B/P

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers FLU-8B/P and FLU-9B/P automatic inflation devices.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

DOD-STD-2101 - Classification of Characteristics

(Copies of this publication are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

30003-849AS103 - Battery, Dry Storage

30003-849AS153 - Inflation Device, Automatic FLU-8B/P

30003-849AS221 - Inflation Device, Automatic FLU-9B/P

(Copies of these publications are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this information using ASSIST at https://assist.dla.mil.

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2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this publication are available online at https://webstore.ansi.org/.)

ASTM INTERNATIONAL

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of this publication are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design and construction. The FLU-8B/P inflation device shall be manufactured in accordance with Drawing 30003-849AS153. The FLU-9B/P inflation device shall be manufactured in accordance with Drawing 30003-849AS221.

3.3 Materials. All materials used in the manufacture of the inflation devices shall conform to the standards established at technical service release (see 6.2). Any divergence will require technical evaluation and potential design certification testing.

3.4 Primary components. For the purposes of this specification, the explosive components are defined as primary components (see 4.5.1). Only primary components from a single lot shall be used in a production lot of inflation devices. One primary component production lot may be used in more than one inflation device production lot.

3.4.1 Bridgewire resistance. The bridgewire resistance of each primary component shall meet the requirements specified in Drawing 30003-849AS153 for FLU-8B/P and Drawing 30003-849AS221 for FLU-9B/P when tested in accordance with 4.5.3.

3.4.2 Radiographic examination. The primary components shall be free of imperfections when examined in accordance with 4.5.7.

3.4.3 Ballistic. The primary components shall have been ballistically tested and accepted in accordance with the contractor's acceptance procedure prior to installation in the inflation device (see 6.2).

3.4.4 Primers. Primers shall have been manufactured within 24 months of the inflation device manufacturing date in accordance with 4.5.8.

3.5 Battery. Batteries shall be in accordance with Drawing 30003-849AS103 and shall have been manufactured within 12 months of the delivery date. Batteries shall be included, but not installed, with inflators delivered for first article inspection and lot acceptance inspection.

3.6 Elastomeric materials. Elastomeric materials shall be within the shelf life determined by the manufacturer at the time of installation in the inflation device.

3.7 Sensor assembly circuit function. Each sensor assembly circuitry shall be electrically tested and accepted in accordance with the acceptance test procedures (see 6.2).

3.8 Leakage.

3.8.1 Leakage at 40 pounds per square inch gauge (psig). The inflation devices shall not show any evidence of leakage when tested in accordance with 4.5.12.1.

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3.8.2 Leakage at 10 psig. The inflation devices shall not show any evidence of leakage when tested in accordance with 4.5.12.2.

3.9 Manual operation.

3.9.1 Lever operation without CO_2 cylinder. Each inflation device shall fully stroke the piercing pin when tested in accordance with 4.5.13.1.

3.9.2 Lever operation with CO_2 cylinder. Sample inflation devices shall fully stroke the piercing pin, perforate the CO_2 cylinder, and release the CO_2 gas when tested in accordance with 4.5.13.2.

3.10 Bleed circuit resistance. The bleed circuit resistance of each inflation device shall be accepted in accordance with the acceptance test procedures (see 6.2).

3.11 Visual examination. Each inflation device shall be in accordance with Drawing 30003-849AS153 or Drawing 30003-849AS221. Each inflation device shall be free of external damage; defects such as cracks, dents, corrosion, illegible or incomplete markings; or missing or misassembled parts when inspected in accordance with 4.5.6.

3.12 Radiographic examination. The inflation devices shall be free of imperfections when examined in accordance with 4.5.7.

3.13 Battery voltage. Batteries installed in inflation devices shall have a minimum voltage of 12.0 volts when measured in accordance with 4.5.9.

3.14 Environmental requirements.

3.14.1 Temperature-shock/humidity/altitude (TSHA). The inflation devices with batteries installed shall not inadvertently function, or otherwise be adversely affected, when tested in accordance with 4.5.11.1.

3.14.2 Vibration. The inflation devices with batteries installed shall not inadvertently function, or otherwise be adversely affected, when tested in accordance with 4.5.11.2.

3.14.3 Rain. The inflation devices with batteries installed shall not inadvertently function, or otherwise be adversely affected, when tested in accordance with 4.5.11.3.

3.14.4 No fire sensitivity. The inflation devices with batteries installed shall not inadvertently function, or otherwise be adversely affected, when tested in accordance with 4.5.11.4.

3.15 Automatic operation. Inflation devices shall meet the following requirements when tested in accordance with 4.5.15.

3.15.1 Function time. Function time is the time from immersion of the inflation device in water to start of release of CO_2 gas. The function time shall be less than 5 seconds after immersion in water in accordance with 4.5.15.1.

3.15.2 Gas release. The CO_2 gas shall be fully released. The CO_2 gas shall not pass through any part of the inflation device other than the gas passages in the outlet port when tested in accordance with 4.5.15.2.

3.15.3 Piercing pin. The piercing pin of the inflation device shall fully stroke and remain in the fully stroked position after automatic inflation in accordance with 4.5.15.3.

3.15.4 Integrity. No part of the inflation device shall rupture or fragment when inspected in accordance with 4.5.15.4.

3.15.5 Post firing inspection.

3.15.5.1 Packing loop release. The packing loop of the FLU-8B/P shall be released and be free to be removed from the inflation device when inspected in accordance with 4.5.15.5.

3.15.5.2 Zipper release. The FLU-9B/P zipper release mechanism shall be free to release when inspected in accordance with 4.5.15.6.

3.16 Lanyard release. The FLU-9B/P lanyard shall not break or fail when a tensile load of 70 pounds is applied to the lanyard when tested in accordance with 4.5.10.

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3.17 Workmanship. Inflation devices shall be constructed and finished in a manner to ensure compliance with all requirements of this specification and associated drawings. Particular attention should be directed to dimensions, finishes, sealing, and assembly operations when examined in accordance with 4.5.6.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Production inspection (see 4.3).

c. Lot acceptance inspection (see 4.4).

4.2 First article inspection. Unless otherwise specified (see 6.2), an acceptable first article sample of 35 FLU-8B/P or FLU-9B/P inflation devices and 70 batteries shall be prepared using the same methods and procedures proposed for the production lot. First article is for the purpose of determining that the contractor, prior to starting production, is capable of producing items that comply with the technical requirements of the contract. Thirty of these inflation devices shall be expended in the tests listed in table I and five shall be retained for investigative purposes. A first article sample acceptable for environmental and functional testing shall be defined as a sample that has met the requirements of 3.11 through 3.15 and 3.17. First article sample acceptance shall consist of the tests shown in table I in the order specified. Failure of any cartridge to comply with the requirements of table I shall be cause for rejection of the first article lot represented.

TABLE I. First article inspections.

Test/Inspection	Requirement Paragraph	Procedure Paragraph	Group A	Group B	Group C	Group D	Total
Visual	3.11, 3.17	4.5.6	18	6	6	5	35
Radiographic	3.12	4.5.7	18	6	6	5	35
TSHA	3.14.1	4.5.11.1	18	-	-	-	18
Vibration	3.14.2	4.5.11.2	-	6	-	-	6
Rain	3.14.3	4.5.11.3	-	-	6	-	6
Radiographic	3.12	4.5.7	-	6	-	-	6
Battery Voltage	3.13	4.5.9	18	6	6	-	30
No Fire Sensitivity	3.14.4	4.5.11.4	18	6	6	-	30
Function at -20 °F	3.15	4.5.15.1	6	2	2	-	10
Function at 70 °F	3.15	4.5.15.1	6	2	2	-	10
Function at 120 °F	3.15	4.5.15.1	6	2	2	-	10
Retained Sample	-	-	-	-	-	5	5

4.3 Production inspection. All production inflation devices shall be inspected and screened for the defects specified in table II. Any inflation device failing to meet the requirements of each test and examination in table II shall be removed from the production lot.

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TABLE II. Production inspections.

Item	Nature of Test	Requirement	Test Method	Classification (DOD-STD-2101)
Primary Component	Single Lot Verification	3.4	4.4	(M101)
Primary Component	Bridgewire Resistance	3.4.1	4.5.3	(M102)
Primary Component	Radiographic Examination	3.4.2	4.5.4	(M103)
Primary Component	Ballistic	3.4.3	-	(C1)
Primers	Age	3.4.4	4.5.8	(M114)
Battery	Age	3.5	4.5.9	(M104)
Elastomeric Materials	Age	3.6	4.5.2	(M105)
Sensor Assembly	Function	3.7	4.5.5	(C2)
Inflation Device	Leakage	3.8	4.5.12	(M107)
Inflation Device	Bleed Circuit Resistance	3.10	4.5.14	(M109)
Inflation Device	Visual Examination	3.11	4.5.6	(M110)
Inflation Device	Radiographic Examination	3.12	4.5.7	(M111)
Inflation Device	Battery Voltage	3.13	4.5.9	(M112)

4.4 Lot acceptance inspection. Lot acceptance inspection shall consist of the tests and examinations specified in table III. Failure of any sample to comply with the requirements specified in table III shall be cause for rejection of the lot represented.

TABLE III. Lot acceptance inspections.

Item	Nature of Test	Requirement	Test Method	Classification (DOD-STD-2101)
FLU-9 Lanyard Assembly	Tensile Test	3.16	4.5.10	(M106)
Inflation Device	Battery Voltage	3.13	4.5.9	(M112)
Inflation Device	No Fire Sensitivity	3.14.4	4.5.11.4	(M113)
Inflation Device	Lever Operation without Cylinder	3.9.1	4.5.13.1	(M108)
Inflation Device	Lever Operation with Cylinder	3.9.2	4.5.13.2	(M109)
Inflation Device	Battery Voltage	3.13	4.5.9	(M112)
Inflation Device	Automatic Operation	3.15	4.5.15	(C3)

4.4.1 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with ANSI/ASQ Z1.4, General Inspection Level II. The lot size shall be between 26 and 1,200 inflation devices. Two inflation devices, in addition to the sample quantity, shall be retained for investigative purposes.

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4.5 Inspections and tests.

4.5.1 Primary components. Use of a single lot of primary components shall be verified by inspection of the materials used for each first article and production lot.

4.5.2 Elastomeric materials. The age of elastomeric materials shall be verified by visual inspection of the manufacture date.

4.5.3 Bridgewire resistance. The bridgewire resistance shall be measured with a test circuit that limits the bridge current to 5 milliamperes, maximum. An instrument that is known to be accurate within 1 percent in the range of resistance specified shall be used to determine whether or not the requirement is met. Any primary component not meeting the requirements of 3.4.1 shall be rejected.

4.5.4 Primary radiographic examination. Each primary component shall be radiographically examined in accordance with ASTM E1742/E1742M; any observable imperfections in assembly shall be cause for rejection. The primary components shall be positioned on their sides for the most revealing exposure with the plane of the pins perpendicular to the x-ray source. All primary components shall be identified utilizing a unique numbering system prior to examination. The primary components shall be arranged sequentially on tray boards utilizing this system. Each radiographic plate shall carry a permanent means enabling identification of individual primer components displayed thereon. The radiograph identification shall include the complete lot number and the span of numbers displayed. Discontinuities in the numbering sequence shall be identified on the x-ray. Defective primary components shall not be used in the inflator production lot. Primary components shall be left on trays or boards until after the x-rays have been reviewed and the defects identified to facilitate removal of defectives.

4.5.5 Sensor assembly functional test. Each sensor assembly circuit shall be electrically tested and accepted per the contractor's manufacturing and acceptance test procedures. The firing circuit shall be able to meet the automatic operation requirements of 3.15. Any changes in the procedure shall be subject to review and approval by the Technical Authority (see 6.2 and 6.5.3).

4.5.6 Visual examination. Inflation devices shall be visually examined and meet the requirements of 3.11 and 3.17.

4.5.7 Radiographic examination. Each inflation device shall be radiographically examined in accordance with ASTM E1742/E1742M; any observable imperfections in assembly shall be cause for rejection. The inflation devices shall be positioned on their sides for the most revealing exposure with the plane of the lever perpendicular to the x-ray source. All inflation devices shall be identified with serial numbers prior to examination. The inflation device serial numbers shall be in consecutive order beginning with the lowest number in each production lot. The inflation devices shall be arranged on tray boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the inflation devices displayed thereon. The radiograph identification shall include the complete lot number and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the x-ray. Defective inflation devices found by the vendor shall be clearly identified on the x-ray and those defectives removed from the production lot.

4.5.8 Primers. Primer age shall be verified by inspection of the manufacturer's date.

4.5.9 Battery. Battery age shall be verified by inspection of the manufacture date. Batteries shall be installed in inflation devices for the battery voltage test. The voltage of each inflation device shall be tested per the contractor's manufacturing or acceptance test procedures. Any changes in the procedure shall be subject to review and approval by the Technical Authority (see 6.2 and 6.5.3). Batteries that do not meet the requirements of 3.13 shall be replaced.

4.5.10 Lanyard tensile test. The FLU-9B/P inflation devices with lanyards installed shall be installed in a tensile tester. The lanyard shall be dry and not twisted more than one half turn. The tensile load shall slowly be applied until 70 pounds of force is applied between the knob and the cam lever. The lanyards shall meet the requirements of 3.16. The test shall only be performed on the lot acceptance test sample.

4.5.11 Environmental tests.

4.5.11.1 Temperature-shock/humidity/altitude. The fully assembled inflation devices including batteries and CO_2 cylinders shall be supported on screen trays so that all areas are exposed to the prescribed atmospheric conditions at all times. The D port in the inflation devices shall be plugged to preclude intrusion of moisture into the devices. The cycling schedule shall be as specified in table IV.

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TABLE IV. TSHA cycling schedule.

Monday	0800	Place test items in a chamber maintained at +70 °F at 50 percent relative humidity (RH).
	1200	Raise chamber temperature to +120 °F and the RH to 95 percent. The chamber temperature shall reach +120 °F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -20 °F at a pressure altitude of 70,000 feet (0.65 psi).
Tuesday	0800	Remove test items from above chamber and immediately place in a chamber maintained at +70 °F at 50 percent RH.
	1200	Remove test items from above chamber and immediately place in a chamber maintained at -20 °F at a pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove test items from above chamber and immediately place in a chamber maintained at +120 °F at 95 percent RH.
Wednesday	0800	Reduce chamber temperature to +70 °F at 50 percent RH. The chamber temperature shall reach +70 °F at 50 percent RH not later than 0900.
	1200	Raise chamber temperature to +120 °F at 95 percent RH. The chamber temperature shall reach +120 °F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -20 °F at a pressure altitude of 70,000 feet (0.65 psi).
Thursday	0800	Remove test items from above chamber and immediately place in a chamber maintained at +70 °F at 50 percent RH.
	1200	Remove test items from the above chamber and immediately place in a chamber maintained at -20 °F at a pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove test items from above chamber and immediately place in a chamber maintained at +120 °F at 95 percent RH.
Friday	0800	Reduce chamber temperature to +70 °F at 50 percent RH. The chamber temperature shall reach +70 °F at 50 percent RH not later than 0900.
	1200	Raise chamber temperature to +120 °F at 95 percent RH. The chamber temperature shall reach +120 °F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -20 °F at standard ambient pressure.

This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of +120 °F at 95 percent RH. The inflation devices shall not inadvertently fire or incur damage during the test.

4.5.11.2 Vibration. The inflation devices shall be equally divided for vibration at -20 °F, 70 °F, and 120 °F. The inflation devices without battery or CO_2 cylinder shall be mounted on a fixture that allows vibration in three mutually perpendicular axes. In each axis, a resonance survey shall be conducted and the resonance points noted (see 6.5.2). A vibration dwell shall be conducted for 30 minutes at each of the four most severe resonance points. The dwell shall be peaked in case the resonant point shifts. After resonant dwell, the inflation devices shall be

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vibration cycled for the time specified in table V. The vibration cycling shall be sinusoidal vibration to the levels shown on figure 1. The frequency shall be logarithmically swept from 5 to 2,000 hertz and return to 5 hertz in 20 minutes.

TABLE V. Vibration test schedule/axis.

Resonances	0	1	2	3	4
Total Dwell Time (minutes)	0	30	60	90	120
Total Cycling Time (minutes)	180	150	120	90	60

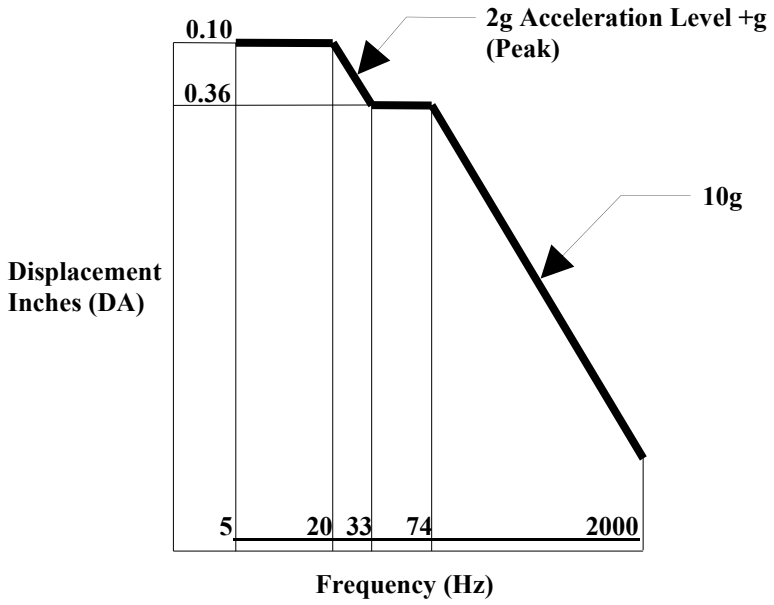


FIGURE 1. Vibration test levels.

4.5.11.3 Rain. The fully assembled inflation devices, including batteries and CO_2 cylinders, shall be placed on plastic screen trays to allow air to circulate freely, water to drain easily, and to expose all surfaces to the environment. The D port in the inflation devices shall be plugged to preclude intrusion of water into the devices. The rainfall rate shall be adjusted to 4±1 inches per hour and wind velocity to 40±4 miles per hour. The rain water conductivity shall be 350±35 micromhos/centimeter (see 6.5.1). These conditions shall be maintained for 30 minutes. The tray shall then be rotated 90 degrees and the test repeated. The tray shall be rotated and the test repeated until all four sides have been tested. The inflation devices shall then be dried at 120 °F for 24 hours. Evidence of inadvertent function shall be cause for rejection.

4.5.11.4 No fire sensitivity. The inflation devices shall be temperature conditioned at 70±5 °F for a minimum of 4 hours. To simulate water conductivity, a 160-kiloohm±10 percent resistor shall be connected between the sensor pin and one of the sensor assembly mounting screws for a minimum of 60 seconds.

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4.5.12 Leakage. Leakage tests shall be performed on inflation devices without shear wire or batteries installed. The test shall be performed on the entire production lot.

4.5.12.1 Leakage at 40 psig. A discharged CO_2 cylinder or equivalent test fitting shall be inserted into the inflation device with batteries removed. The lever shall be in the actuated position during this inspection. The actuated position is considered to be the position of the lever after the piercing pin has punctured the CO_2 cylinder. A pressure of 40±2.0 psig shall be applied at the outlet of the inflation device for a minimum of 10 seconds. While the pressure is applied, the inflation device shall be observed for leakage. Any detectable pressure drop shall be cause for rejection of the inflation device. Upon completion of this inspection, the inflation device shall be subjected to the inspection of 4.5.12.2.

4.5.12.2 Leakage at 10 psig. A discharged CO_2 cylinder or equivalent test fitting shall be inserted into the inflation device with batteries removed. The lever shall be in the actuated position. A pressure of 10±1.0 psig shall be applied at the outlet of the inflation device for a minimum of 10 seconds. While the pressure is applied, the lever shall be slowly moved through at least one complete normal operation and the inflation device shall be observed for leakage. Any indication of leakage shall be cause for rejection of the inflation device.

4.5.13 Manual operation.

4.5.13.1 Lever operation without CO_2 cylinder. The lever of the FLU-8B/P and FLU-9B/P shall not be secured by a shear wire. For the weighted test, the inflation device shall be oriented vertically with the open CO_2 cylinder port up. A weight of 3+0/-0.5 pounds shall be attached to the free end of the lever and the lever shall be observed for complete rotation. This test shall be applied to 4 percent of the lot quantity rounded to the nearest whole number or the lot acceptance test sample size, whichever is less. That quantity shall be taken from the lot acceptance test sample. Any inflation device where the lever does not completely rotate shall be cause for rejection of the lot. The sample shall then be tested in accordance with 4.5.13.2. Manual activation of the lever without the 3-pound weight shall be done on the entire lot.

4.5.13.2 Lever operation with CO_2 cylinder. A CO_2 filled cylinder shall be fully installed in the inflation device and hand tightened. The CO_2 cylinder shall then be removed and its diaphragm face examined for any evidence of puncture or indentations by the piercing pin. Upon completion of this inspection, the cylinder shall be reinstalled in the inflation device. The inflation device shall be oriented vertically with the CO_2 cylinder up. A 20-pound maximum weight shall be attached to the free end of the lever. The piercing pin shall advance to its maximum point of travel, pierce the diaphragm of the CO_2 cylinder, and release the CO_2 gas. Any inflation device that does not meet the requirements of 3.9.2 shall be cause for rejection of the lot.

4.5.14 Bleed circuit resistance. The bleed circuit resistance shall be tested in accordance with the contractor's manufacturing or acceptance test procedure. Any changes in the procedure or resistance value shall be subject to review and approval by the Technical Authority (see 6.2 and 6.5.3). If the resistance reading is not in accordance with the approved procedure, the inflation device shall be rejected.

4.5.15 Automatic operation tests.

4.5.15.1 Functional test. The fully assembled inflation devices shall be equally divided for temperature conditioning at -20±5 °F, 70±5 °F, and 120±5 °F for a minimum of 4 hours. The inflation devices shall be functioned within 3 minutes after removal from temperature conditioning. The inflation devices shall be functioned by immersion in water at 32+5/-0 °F and 67±5 micromho/centimeter conductivity (see 6.5.1). The time from immersion in water to start of release of CO_2 gas shall be measured.

4.5.15.2 Gas release. The functioned inflation devices shall be inspected for complete release of gas from the CO_2 cylinder through the outlet port (see 3.9.2 and 3.15.2).

4.5.15.3 Piercing pin. The functioned inflation devices shall be inspected for complete stroke of the piercing pin (see 3.15.3). Verification of full stroke shall be confirmed by inspection of the inflation device visual indicator, verifying locking of the piercing pin in the actuated position, or inspection of the CO_2 cylinder diaphragm.

4.5.15.4 Integrity. The functioned inflation devices shall be inspected for integrity (see 3.15.4).

4.5.15.5 Packing loop release. The functioned FLU-8B/P inflation devices shall be inspected for released packing loop (see 3.15.5.1).

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4.5.15.6 Zipper release. The functioned FLU-9B/P inflation devices shall be inspected for zipper release operation (see 3.15.5.2).

4.5.16 Acceptance criteria. The total number of allowed defective units in a first article sample or lot acceptance sample as a result of tests and inspections of 3.8 through 3.15 shall be zero. Any defects shall result in rejection of the lot represented. Screening of defects as a corrective action and the method of screening shall be subject to approval by the Technical Authority (see 6.5.3).

4.5.17 Retest. There shall be no retests.

4.5.18 Test failure. If a test failure is attributable to an assignable cause, excluding the inflation devices, the original individual test results shall be discarded and the complete test repeated.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The FLU-8B/P and FLU-9B/P inflation devices are designed to inflate life preservers either manually or automatically by opening a CO_2 cylinder and releasing the gas into the life preserver through an outlet. These inflation devices were designed for military use and have no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Designation of inflation device (FLU-8B/P and FLU-9B/P).

c. When first article inspection is not required (see 3.1 and 4.2).

d. Materials required as established by the technical service release (see 3.3).

e. Requirements for Technical Authority approval of the acceptance test procedures (see 3.4.3, 3.7, 3.10, and 6.5.3).

f. Production lot size and test activity (see 4.4).

g. The lot acceptance test sample size if other than specified (see 4.4.1).

h. Packaging requirements and marking (see 5.1 and 6.4).

i. Items of data required for each first article and production lot (see 6.3).

j. That the safety precaution requirements of DoDM 4145.26-M, DoD Contractor's Safety Manual for Ammunition and Explosives, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of OP 5, Ammunition and Explosives Ashore, are applicable.

6.3 Data. For the information of contractors and contracting officers, any of the data specified in 6.3.1, applicable documents listed in section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government unless specified (see 6.2).

6.3.1 Standard deliverables. In addition to the delivery of inflation devices, the contract may require delivery of the following:

a. Radiographic plates for FLU-8B/P or FLU-9B/P inflators.

b. Radiographic review report.

c. Radiographic sample plates and techniques.

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d. Requests for waiver/deviation and engineering change proposal (ECP).

e. Test and inspection report, including acceptance test plan if contractor tested.

f. Ammunition data cards.

g. Failure analysis report and corrective action plan.

6.4 Contract packaging and marking. The following packing and marking requirements should be specified in the contract.

6.4.1 Preservation and packing. Preservation/packing should be in accordance with the 49 CFR 171-178. The contractor is responsible for performance-oriented packing certification and documentation in accordance with 49 CFR 171-178. Level A preservation and packing should be used for packaging of all production lots. Inflation devices should be packaged in an inner container in accordance with SPI-PHST-283. Inner containers should be packed in the outer container in accordance with SPI-PHST-283. For an explanation of terms, see 6.4.2.

6.4.2 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.28, when applicable, is used to clarify terms.

6.4.3 Markings.

6.4.3.1 Special markings. Marking of exterior containers should be in accordance with 49 CFR 171-178. Markings should be applied by the contractor or their agent.

6.4.3.2 Normal markings. Unless otherwise specified (see 6.2), the marking information on unit packages and shipping containers should be as specified on SPI-PHST-283. The specified marking information should be applied to the containers in accordance with the applicable provisions of MIL-STD-129. Lot number should be in accordance with MIL-STD-1168.

6.5 Definitions.

6.5.1 Micromho/centimeter. Micromho/centimeter is defined as the conductance in 1 centimeter of water through which 1 millionth (0.000001) ampere flows when the potential is 1 volt.

6.5.2 Resonance. The measured vibration response (output) is at least twice as strong as the vibration being applied (input).

6.5.3 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05T may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.6 Subject term (key word) listing.

Bleed circuit
Bridgewire
Elastomeric material
Inflator
Life preservers
Primary component
Sensor circuit

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2024-009)

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